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Unit – 3 $\rightsquigarrow$ Probability Distribution

Method 1 $\rightsquigarrow$ Binomial Distribution

Introduction

- In this chapter we shall study some of the probability distribution that figure most prominently in statistical theory and application. We shall also study their parameters. We shall introduce number of discrete probability distribution that have been successfully applied in a wide variety of decision situations.
- The purpose of this chapter is to show the types of situations in which these distributions can be applied.
- Probability function of discrete random variable is known as probability mass function (P.M.F.) and probability function of continuous random variable is known as probability density function (P.D.F.).
- Some special probability distributions:
 - (1) Binomial distribution (P.M.F.)
 - (2) Poisson distribution (P.M.F.)
 - (3) Normal distribution (P.D.F.)
 - (4) Exponential distribution (P.D.F.)

Bernoulli Trials

- Suppose a random experiment has two possible outcomes, which are
 - (1) Success (S)
 - (2) Failure (F)
- If the probability $p(0 < p < 1)$ of getting success at each of the n trials of this experiment is constant, then the trials are known as Bernoulli trials.

Binomial Distribution

- A random experiment consists of n Bernoulli trials such that the trials are independent.
- Each trial results in only two possible outcomes, labeled as success and failure.
- The probability of success in each trial remains constant.
- The random variable X that equals the number of trials that results in a success is a binomial random variable with parameters $0 < p < 1$, $q = 1 - p$ and $n = 1, 2, 3, \dots$

Unit 3 Probability Distribution

→ The probability mass function of X is

$$P(X = x) = \binom{n}{x} p^x q^{n-x} ; x = 0, 1, 2, \dots, n$$

→ For Examples:

- (1) Number of defective bolts in a box containing n bolts.
- (2) Number of post-graduates in a group of n people.
- (3) Number of oil wells yielding natural gas in a group of n wells test drilled.
- (4) In the next 20 births at a hospital. Let X = the number of female births.
- (5) Flip a coin 10 times. Let X = number of heads obtained.

Properties of Binomial Distribution

- The mean of binomial distribution is defined as $\mu = E(X) = np$.
- The variance of binomial distribution is defined as $V(X) = npq$.
- The standard deviation of binomial distribution is defined as $\sigma = \sqrt{npq}$.

Example of Method-1: Binomial Distribution

C	1	For the binomial distribution with $n = 20, p = 0.35$. Find mean, variance and standard deviation. Answer: Mean = 7, Variance = 4.55, Standard Deviation = 2.1331												
C	2	Obtain the binomial distribution function for which mean is 10 and variance is 5. Answer: $P(X = x) = \binom{20}{x} (0.5)^x (0.5)^{20-x} ; x = 0, 1, 2, \dots, 20$.												
H	3	Find the binomial distribution for $n = 4$ and $p = 0.3$. Answer: <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>P(X)</td><td>0.2401</td><td>0.4116</td><td>0.2646</td><td>0.0756</td><td>0.0081</td></tr></table>	X	0	1	2	3	4	P(X)	0.2401	0.4116	0.2646	0.0756	0.0081
X	0	1	2	3	4									
P(X)	0.2401	0.4116	0.2646	0.0756	0.0081									

Unit 3 Probability Distribution

H	4	If the probability of a defective bolt is 0.1, find mean and standard deviation of the distribution of defective bolts in a total of 400. Answer: Mean = 40, Standard Deviation = 6
C	5	12% of the tablets produced by a tablet machine are defective. What is the probability that out of a random sample of 20 tablets produced by the machine, 5 are defective? Answer: 0.0567
H	6	If 3 of 12 car drivers do not carry driving license, what is the probability that a traffic inspector who randomly checks 3 car drivers, will catch 1 for not carrying driving license. Answer: $\frac{27}{64}$
T	7	Find the probability that in a family of 4 children there will be at least 1 boy. Assume that the probability of a male birth is 0.5. Answer: 0.9375
H	8	20% of the bulbs produced are defective. Find probability that at most 2 bulbs out of 4 bulbs are defective. Answer: 0.9728
H	9	A university warehouse has received a shipment of 25 printers, of which 10 are laser printers and 15 are inkjet models. If 6 of these 25 are selected at random to be checked by a particular technician, what is the probability that exactly 3 of those selected are laser printers (so that the other 3 are inkjets)? Answer: 0.2765
H	10	What are the properties of Binomial Distribution? The average percentage of failure in a certain examination is 40. What is the probability that out of a group of 6 candidates, at least 4 passed in examination? Answer: 0.5443

Unit 3 Probability Distribution

C	11	Find probability of getting a sum of 7 at least once in 3 tosses of a pair of dice. Answer: $\frac{91}{216}$
H	12	Each sample of water has a 10% chance of containing a particular organic pollutant. Assume that the samples are independent with regard to the presence of the pollutant. Find the probability that in the next 18 samples, at least 4 samples contain the pollutant. Answer: 0.0982
H	13	Ten coins are thrown simultaneously. Find the probability of getting at least seven heads. Answer: 0.1719
C	14	The probability that India wins a cricket test match against Australia is given to be $\frac{1}{3}$. If India and Australia play 3 test matches, what is the probability (a) India will lose all the three test matches? (b) India will win at least one test match? Answer: (a) 0.2963, (b) 0.7037
T	15	Assume that on average one telephone number out of fifteen called between 1 p.m. and 2 p.m. on weekdays is busy. What is the probability that, if 6 randomly selected telephone numbers were called, (a) not more than three? (b) at least three, of them would be busy? Answer: (a) 0.9994, (b) 0.0006
H	16	The probability that one of the ten telephone lines is busy at an instant is 0.6. (a) What is the probability that 5 of the lines are busy? (b) What is the probability that all the lines are busy? Answer: (a) 0.2006, (b) 0.0060

Unit 3 Probability Distribution

H	17	The probability that in a university, a student will be a post-graduate is 0.6. Determine probability that out of 8 students none, two and at least two will be post-graduate. Answer: 0.0007, 0.0413, 0.9914
H	18	If 20% of the bolts produced by a machine are defective, determine the probability that out of 4 bolts chosen at random, one, zero and less than 2, bolts will be defective. Answer: 0.4096, 0.4096, 0.8192
H	19	Probability of man hitting a target is $\frac{1}{3}$. If he fires 6 times, what is the probability of hitting at most 5 times, at least 5 times, and exactly one? Answer: 0.9986, 0.0179, 0.2634
H	20	The probability that an infection is cured by a particular antibiotic drug within 5 days is 0.75. Suppose 4 patients are treated by this antibiotic drug. What is the probability that no patient, exactly two patients, and at least two patients, are cured? Answer: 0.0039, 0.2109, 0.9492
C	21	By assuming equal probabilities for boys and girls. Out of 800 families with 4 children each, how many would you expect to have (a) 2 boys and 2 girls? (2) at least 1 boy? (3) at most 2 girls? (4) no girl? Answer: (a) 300, (b) 750, (c) 550, (d) 50
H	22	Out of 2000 families with 4 children each, how many would you expect to have at least 1 boy, 2 boys, 1 OR 2 girls, and no girls? Assume equal probabilities for boys and girls. Answer: 1875, 750, 1250, 125

Unit 3 Probability Distribution

C	23	Find the probability that in five tosses of a fair die, 3 will appear twice, at most once, at least two times. Answer: $\frac{625}{3888}$, $\frac{3125}{3888}$, $\frac{763}{3888}$
H	24	A dice is thrown 6 times getting an odd number of successes. Find probability of five successes, at least five successes and at most five successes. Answer: $\frac{3}{32}$, $\frac{7}{64}$, $\frac{63}{64}$
T	25	A multiple-choice test consists of 8 questions with 3 answers to each question (of which only one is correct). A student answers each question by rolling a balanced dice & checking the first answer if he gets 1 or 2, the second answer if he gets 3 or 4 & the third answer if he gets 5 or 6. To get a distinction, the student must secure at least 75% correct answers. If there is no negative marking, what is the probability that the student secures a distinction? Answer: $P(X \geq 6) = 0.0197$

Unit 3 Probability Distribution

Method 2 $\rightsquigarrow$ Poisson Distribution

Poisson Distribution

- A discrete random variable X is said to follow Poisson distribution if it assumes only non-negative values.
- Its probability mass function is given by

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, 3, \dots$$

Where, λ = mean of the poisson distribution

- For Examples:
 - (1) Number of telephone calls per minute at a switchboard.
 - (2) Number of cars passing a certain point in one minute.
 - (3) Number of printing mistakes per page in a large text.
 - (4) Number of persons born blind per year in a large city.

Properties of Poisson Distribution

- The random variable X should be discrete.
- The number of trials n is very large.
- The probability of success p is very small (very close to zero).
- The occurrences are rare.
- The mean and variance of the Poisson distribution with parameter λ are defined as
mean $\mu = E(X) = \lambda = np$ & variance $V(X) = \sigma^2 = \lambda$.

Example of Method-2: Poisson Distribution

C	1	For Poisson variant X, if $P(X = 3) = P(X = 4)$, then find $P(X = 0)$. Answer: $P(X = 0) = e^{-4}$
H	2	For Poisson variant X, if $P(X = 1) = P(X = 2)$, then find mean and standard deviation of this distribution. Also, find $P(X = 3)$. Answer: Mean = 2, SD = $\sqrt{2}$, $P(X = 3) = 0.1804$

Unit 3 Probability Distribution

C	3	The number of page requests that arrive at a Web server is a Poisson random variable. Its probability distribution is as follows: <table><tr><td>Number of x requests/sec.</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Probability f(x)</td><td>0.368</td><td>0.368</td><td>0.184</td><td>0.061</td><td>0.015</td><td>0.003</td><td>0.001</td></tr></table> Find the mean and variance of this probability distribution. Answer: 1, 1	Number of x requests/sec.	0	1	2	3	4	5	6	Probability f(x)	0.368	0.368	0.184	0.061	0.015	0.003	0.001
Number of x requests/sec.	0	1	2	3	4	5	6											
Probability f(x)	0.368	0.368	0.184	0.061	0.015	0.003	0.001											
H	4	Assume that the probability that a wafer contains a large particle of contamination is 0.01 and that the wafers are independent; that is, the probability that a wafer contains a large particle is not dependent on the characteristics of any of the other wafers. If 15 wafers are analyzed, what is the probability that no large particles are found? Answer: 0.8607																
H	5	Suppose a book of 585 pages contains 43 typographical errors. If these errors are randomly distributed throughout the book, what is the probability that 10 pages, selected at random, will be free from error? Answer: 0.4795																
T	6	A book contains 100 misprints distributed randomly throughout its 100 pages. What is the probability that a page observed at random contains at least two misprints? Answer: 0.2642																
C	7	In sampling a large number of parts manufactured by a machine, the mean number of defectives in a sample of 20 is 2. Out of 1000 such samples, how many would be expected to contain exactly two defective parts? Answer: 271																
H	8	100 Electric bulbs are found to be defective in a lot of 5000 bulbs. Find the probability that at the most 3 bulbs are defective in a box of 100 bulbs. Answer: $P(X \leq 3) = 0.8571$																

Unit 3 Probability Distribution

C	9	Potholes on a highway can be serious problems. The past experience suggests that there are, on an average, 2 potholes per mile after a certain amount of usage. It is assumed that Poisson process applies to random variable “no. of potholes”. What is the probability that no more than four potholes will occur in a given section of 5 miles? Answer: $P(X \leq 4) = 0.0293$
T	10	In a company, there are 250 workers. The probability of a worker remains absent on any one day is 0.02. Find the probability that on a day, seven workers are absent. Answer: 0.1044
H	11	The number of monthly breakdowns of a computer is a random variable having Poisson distribution with mean 1.8. Find the probability that the computer will function for a month (a) without a breakdown. (b) with at least one breakdown. Answer: (a) 0.1653, (b) 0.8347
H	12	In a bolt manufacturing company, it is found that there is a small chance of $\frac{1}{500}$ for any bolt to be defective. The bolts are supplied in a packed of 20 bolts. Use Poisson distribution to find approximate number of packets containing no defective bolt, containing two defective bolts, in the consignment of 10000 packets. Answer: 9608, 8
H	13	Average number of accidents on any day on a national highway is 1.8. Determine the probability that the number of accidents are at least 1 and at most 1. Answer: 0.8347, 0.4628

Unit 3 Probability Distribution

T	14	The number of flaws in a fiber optic cable follows a Poisson process with an average of 0.6 per 100 feet. (a) Find the probability of exactly 2 flaws in a 200 feet cable. (b) Find the probability of exactly 1 flaw in the first 100 feet and exactly 1 flaw in the second 100 feet. Answer: (a) 0.2169, (b) 0.1084
H	15	If a publisher of nontechnical books takes great pains to ensure that its books are free of typographical errors, so that the probability of any given page containing at least one such error is 0.005 and errors are independent from page to page. What is the probability that one of its 400-page novels will contain (a) exactly one page with errors? (b) at most three pages with errors? Answer: (a) 0.2707, (b) 0.8571
H	16	If the probability that an individual suffers a bad reaction from a certain injection is 0.001. Find the probability that out of 2000 individuals (a) more than 2 individuals (b) exactly 3 individuals will suffer a bad reaction. Answer: (a) 0.3233, (b) 0.1804
C	17	The probability that a person catch corona virus is 0.001. Find the probability that out of 3000 persons exactly 3, more than 2 persons will catch the virus. Answer: 0.2240, 0.5767
T	18	In a certain factory turning out razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packet of 10. Calculate the approximate number of packets containing no defective, one defective, two defective blades in a consignment of 10000 packets. Answer: 9802, 196, 2

Unit 3 Probability Distribution

H	19	A car hire firm has two cars, which are hires out day by day. The number of demands for a car on each day is distributed on a Poisson distribution with mean 1.5. Calculate the proportion of days on which neither car is used and proportion of days on which some demand is refused. ($e^{-1.5} = 0.2231$). Answer: $P(X = 0) = 0.2231$, $1 - P(X \leq 2) = 0.1912$
C	20	Suppose 1% of the items made by machine are defective. In a sample of 100 items find the probability that the sample contains all good, 1 defective and at least 3 defectives. Answer: $P(X = 0) = 0.3679$, $P(X = 1) = 0.3679$ $P(X \geq 3) = 0.0803$

Method 3 $\rightsquigarrow$ Exponential Distribution

Exponential Distribution

- A random variable X is said to have an exponential distribution with parameter $\theta > 0$, if its probability density function is given by

$$f(X = x) = \begin{cases} \theta \cdot e^{-\theta x} & ; x \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

Where, $\theta = \frac{1}{\text{mean}}$ and variance $= \frac{1}{\theta^2}$.

- In exponential distribution we can find the probability as given below:

$$(1) P(X \leq x) = 1 - e^{-\theta x}$$

$$(2) P(X \geq x) = e^{-\theta x}$$

$$(3) P(a \leq X \leq b) = e^{-a\theta} - e^{-b\theta}$$

Properties of Exponential Distribution

- Exponential distribution is used to describe lifespan and waiting times.
- Exponential distribution can be used to describe (waiting) times between Poisson events.

Example of Method-3: Exponential Distribution

C	1	The lifetime T of an alkaline battery is exponentially distributed with $\theta = 0.05$ per hour. What are mean and standard deviation of batteries lifetime? Answer: 20, 20
C	2	A random variable has an exponential distribution with probability density $f(x) = \begin{cases} \frac{1}{5} e^{-\frac{x}{5}} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$ Find (a) $P(X > 5)$, (b) $P(3 \leq X \leq 6)$, (c) Mean, (d) Variance Answer: (a) 0.3679, (b) 0.2476, (c) 5, (d) 25

Unit 3 Probability Distribution

H	3	A random variable has an exponential distribution with probability density $f(x) = \begin{cases} \frac{1}{6} & ; -3 < x < 3 \\ 0 & ; \text{otherwise} \end{cases}$ Find (a) $P(X < 2)$, (b) $P(X \leq 2)$ Answer: (a) $\frac{5}{6}$, (b) $\frac{2}{3}$
T	4	A random variable has an exponential distribution with probability density $f(x) = \begin{cases} ce^{-2x} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$ Find value of the constant c and $P(X > 5)$. Answer: 2, 0.0183
H	5	The lifetime T of an alkaline battery is exponentially distributed with $\theta = 0.02$ per hour. What are mean and variance of batteries lifetime? Answer: 50, 2500
H	6	The time required to repair a machine is exponentially distributed with parameter $\theta = 0.5$ per hour. What is the probability that the repair time exceeds 3 hours? Answer: 0.2231
T	7	Suppose the length of life of an appliance has an exponential distribution with mean 10 years. What is the probability that the average life time of a random sample of the appliances is at least 10.5? Answer: 0.3499
H	8	The time intervals between successive barges passing a certain point on a busy waterway have an exponential distribution with mean 8 minutes. Find the probability that the time interval between two successive barges is less than 5 minutes. Answer: 0.4647

Unit 3 Probability Distribution

C	9	The time between breakdowns of a particular machine follows an exponential distribution with a mean of 17 days. Calculate the probability that a machine breakdown in 15-day period. Answer: 0.5862
C	10	The lifetime T of an alkaline battery is exponentially distributed with $\theta = 0.05$ per hour. (a) What are the probabilities for battery to last between 10 and 15 hours? (b) What are the probabilities for the battery to last more than 20 hours? Answer: (a) 0.1341, (b) 0.3679
H	11	On the average, a certain computer part lasts ten years. The length of time the computer part lasts is exponentially distributed. (a) What is the probability that a computer part lasts more than 7 years? (b) What is the probability that a computer part lasts between 9 and 11 years? Answer: (a) 0.4966, (b) 0.0737
H	12	The mileage which car owners get with a certain kind of radial tire is a random variable having exponential distribution with mean 4000 km. What is the probability that one of these tiers will last (a) at least 2000 km, (b) at most 3000 km. Answer: (a) 0.6065, (b) 0.5276
C	13	In a large corporate computer network, user log-on to the system can be modeled as a Poisson process with a mean of 25 log-on per hours. (a) What is the probability that there are no log-on in an interval of six min.? (b) What is the probability that time until next log-on is between 2 & 3 min.? Answer: (a) 0.0821, (b) 0.1481

Unit 3 Probability Distribution

T	14	Accidents occur with Poisson distribution at an average of 4 per week. (a) Calculate the probability of more than 5 accidents in any one week (b) What is probability that at least two weeks will elapse between accidents? Answer: (a) 0.2149, (b) 0.0003
H	15	The arrival rate of cars at a gas station is 40 customers per hour. (a) What is the probability of having no arrivals in 5 min. interval? (b) What is the probability of having 3 arrivals in 5 min. interval? Answer: (a) 0.0357, (b) 0.2202
T	16	The length of time X to complete a job is exponentially distributed with $E(X) = \mu = \frac{1}{\lambda} = 10$ hours. (a) Compute the probability of job completion between two consecutive jobs exceeding 20 hours. (b) The cost of job completion is given by $C = 4 + 2X + 2X^2$, then find the expected value of C. Answer: (a) 0.1353, (b) 424
H	17	A random variable has an exponential distribution with probability density function given by $f(x) = 3e^{-3x}$; $x > 0$ & $f(x) = 0$; $x \leq 0$. What is the probability that X is not less than 4? Answer: e^{-12}
C	18	The income tax of a man is exponentially distributed with $f(x) = \frac{1}{3}e^{-\left(\frac{x}{3}\right)}$; $x > 0$. What is the probability that his income will exceed Rs. 17000? Assume that the income tax is levied at the rate of 15% on the income above Rs. 15000. Answer: 0.9048

T	19	The life of an electronic component follows exponential distribution with a mean of 4 years. The manufacturer of this component gives a replacement warranty of 3 years. (a) What proportion of components will be replaced in the period of warranty? (b) What is the probability that a randomly selected component will have life within two standard deviations of the mean life? Answer: (a) 0.5276, (b) 0.9502
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Method 4 $\rightsquigarrow$ Normal Distribution

Normal Distribution

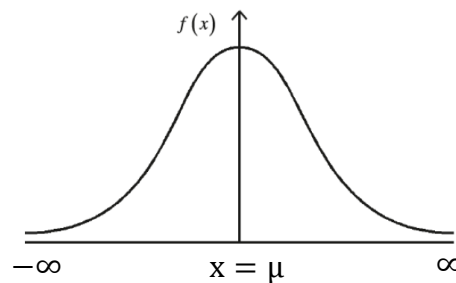
→ A continuous random variable X is said to follow a normal distribution if its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

Where, $-\infty < x < \infty$ & $\sigma > 0$

μ = mean of the distribution

σ = standard deviation of the distribution



→ Here, μ (mean) & σ^2 (variance) are known as parameters of the distribution.

→ If X is a normal random variable with mean μ and standard deviation σ , then we can find the random variable defined as below

$$Z = \frac{X - \mu}{\sigma}$$

→ If $\mu = 0$ and $\sigma = 1$, then Z is known as the standard (standardized) normal variable and probability density function is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}; -\infty < z < \infty$$

→ The distribution of any normal variate X can always be transformed into the distribution of the standard normal variate Z .

$$P(x_1 \leq X \leq x_2) = P\left(\frac{x_1 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{x_2 - \mu}{\sigma}\right) = P(z_1 \leq Z \leq z_2).$$

→ This probability is equal to area under the standard normal curve between the coordinates at $Z = z_1$ and $Z = z_2$.

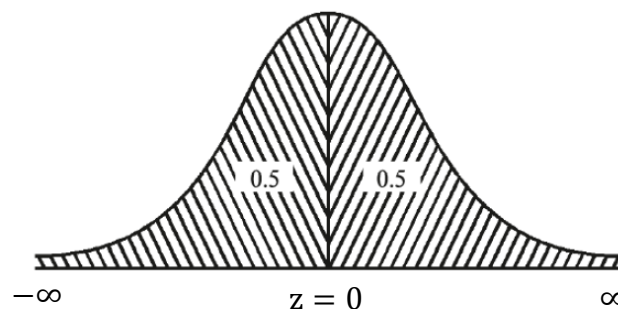
Unit 3 Probability Distribution

Methods for Finding the Probability(area) from Z Table

- Normal curve is a curve of normal density function and it can be seen in figure.
- The probability(area) of shaded region between the curve and x – axis is equal to 1.
- The probability(area) to the right side of $X = \mu$ is 0.5 and is denoted by

$$P(X \geq \mu) = 0.5$$
- The probability(area) to the left side of $X = \mu$ is also 0.5 and is denoted by

$$P(X \leq \mu) = 0.5$$



- Let us discuss some cases which is useful to solve examples.

Important Cases

(1) Area between $Z = 0$ to $Z = z_1 = P(0 \leq Z \leq z_1)$

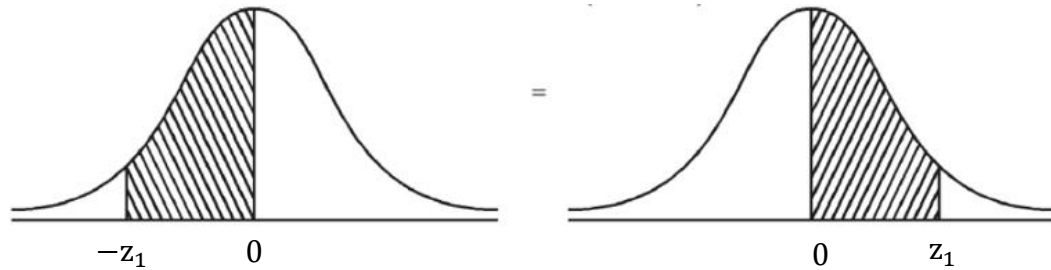
- Find z_1 by using first row and column of Z table to get the answer of $P(0 \leq Z \leq z_1)$.

- For Example:

- Suppose $z_1 = 1.43$. Use below steps to find value at $z = 1.43$.
- Step 1: Split 1.43 as $1.4 + 0.03$.
- Step 2: Find 1.4 in first column and 0.03 in first row.
- Step 3: Intersection of that row and column is answer of $P(0 \leq Z \leq 1.43)$.
- Here, $P(0 \leq Z \leq 1.43) = 0.4236$

- Note:

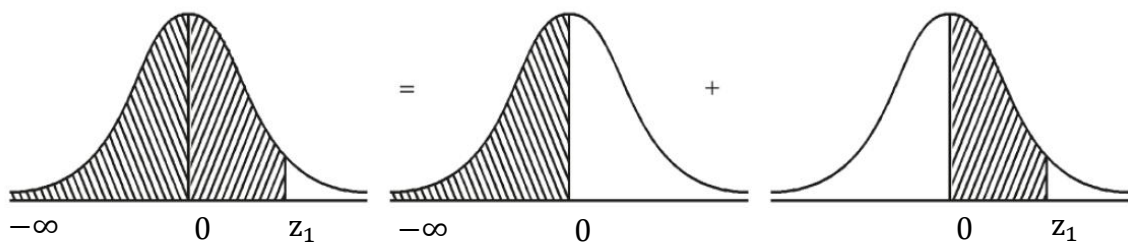
- Area between $Z = -z_1$ to $Z = 0 = P(-z_1 \leq Z \leq 0) = P(0 \leq Z \leq z_1)$ as normal distribution is symmetric about x – axis.
i.e., $P(0 \leq Z \leq 1.43) = P(-1.43 \leq Z \leq 0) = 0.4236$



(2) **Area between $Z = -\infty$ to $Z = z_1 = P(Z \leq z_1)$**

$$\rightarrow P(Z \leq z_1) = P(-\infty \leq Z \leq 0) + P(0 \leq Z \leq z_1)$$

$\rightarrow$ Since, $P(-\infty \leq Z \leq 0) = 0.5$ and $P(0 \leq Z \leq z_1)$ can be found by using case - I.



$\rightarrow$ Note:

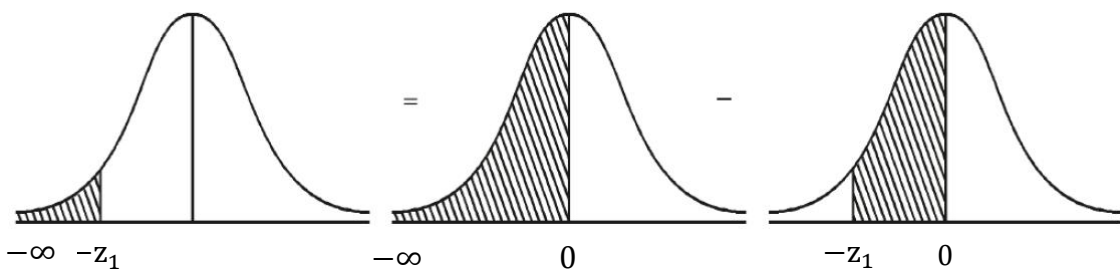
- Area between $Z = -z_1$ to $Z = \infty = P(Z \geq -z_1) = P(Z \leq z_1)$ as normal distribution is symmetric about x - axis.

$$\text{i.e., } P(Z \geq -1.43) = P(Z \leq 1.43) = 0.9236$$

(3) **Area between $Z = -\infty$ to $Z = -z_1 = P(Z \leq -z_1)$**

$$\rightarrow P(Z \leq -z_1) = P(-\infty \leq Z \leq 0) - P(-z_1 \leq Z \leq 0)$$

$\rightarrow$ Since, $P(-\infty \leq Z \leq 0) = 0.5$ and $P(-z_1 \leq Z \leq 0)$ can be found by using case - I.



$\rightarrow$ Note:

- Area between $Z = z_1$ to $Z = \infty = P(Z \geq z_1) = P(Z \leq -z_1)$ as normal distribution is symmetric about x - axis.

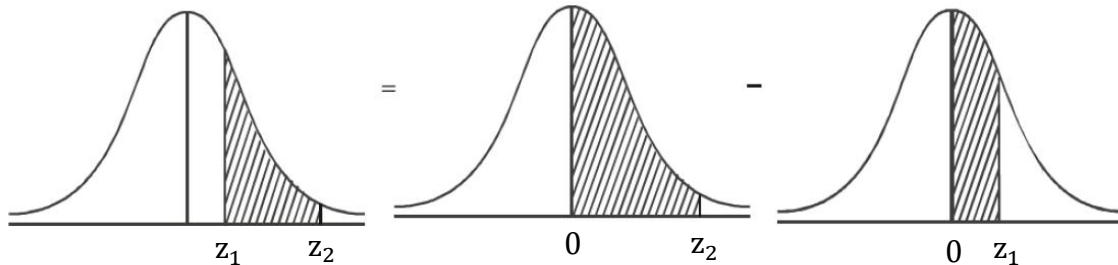
$$\text{i.e., } P(Z \geq 1.43) = P(Z \leq -1.43) = 0.0764$$

Unit 3 Probability Distribution

(4) Area between $Z = z_1$ to $Z = z_2 = P(z_1 \leq Z \leq z_2)$

$$\rightarrow P(z_1 \leq Z \leq z_2) = P(0 \leq Z \leq z_2) - P(0 \leq Z \leq z_1)$$

$\rightarrow$ Since, $P(0 \leq Z \leq z_1)$ and $P(0 \leq Z \leq z_2)$ can be found by using case - I.



$\rightarrow$ Note:

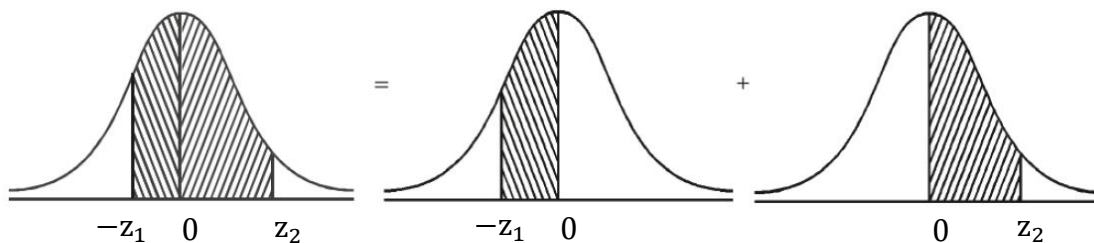
- Area between $Z = -z_2$ to $Z = -z_1 = P(-z_2 \leq Z \leq -z_1) = P(z_1 \leq Z \leq z_2)$ as normal distribution is symmetric about x - axis.

$$\text{i.e., } P(-2 \leq Z \leq -1.43) = P(1.43 \leq Z \leq 2) = 0.0436$$

(5) Area between $Z = -z_1$ to $Z = z_2 = P(-z_1 \leq Z \leq z_2)$

$$\rightarrow P(-z_1 \leq Z \leq z_2) = P(-z_1 \leq Z \leq 0) + P(0 \leq Z \leq z_2)$$

$\rightarrow$ Since, $P(-z_1 \leq Z \leq 0)$ and $P(0 \leq Z \leq z_2)$ can be found by using case - I.



$\rightarrow$ Note:

- Area between $Z = -z_1$ to $Z = z_2 = P(-z_1 \leq Z \leq z_2)$ as normal distribution is symmetric about x - axis.

$$\text{i.e., } P(-1 \leq Z \leq 2) = P(-1 \leq Z \leq 0) + P(0 \leq Z \leq 2)$$

$$= 0.3413 + 0.4772$$

$$= 0.8185$$

Unit 3 Probability Distribution

Properties of Normal Distribution

- The curve is bell shaped and symmetrical about the line $x = \mu$.
- Mean, median and mode of the distribution coincide. i.e. the normal distribution is symmetrical.
- As x increases numerically, $f(x)$ decreases rapidly, the maximum probability occurring at the point $x = \mu$.
- x - axis is an asymptote to the curve.

Importance

- Most of the distributions occurring in practice, e.g., Binomial, Poisson, etc. can be approximated by normal distribution.
- Even if a variable is not normally distributed, it can sometimes be brought to normal form by simple transformation of variable.
- The entire theory of small sample tests is based on the fundamental assumption that the parent populations from which the samples have been drawn follows normal distribution.

Example of Method-4: Normal Distribution

H	1	If mean is 50 and standard deviation is 10, then find $P(50 \leq X \leq 80)$. [$P(0 < z < 3) = 0.4987$] Answer: 0.4987
T	2	If X is a normal variate with a mean of 30 and an SD of 5, find the probability that (i) $26 \leq X \leq 40$, (ii) $X \geq 45$. Answer: (i) 0.7653, (ii) 0.0013
H	3	The breaking strength of cotton fabric is normally distributed with $E(x) = 16$ and $\sigma(x) = 1$. The fabric is said to be good if $x \geq 14$. What is the probability that a fabric chosen at random is good? [$P(0 < z < 2) = 0.4772$] Answer: 0.9773

Unit 3 Probability Distribution

C	4	If X is normally distributed and the mean of X is 12 and the SD is 4. Find out the probability of the following. (a) $X \geq 20$, (b) $X \leq 20$, (c) $0 \leq X \leq 12$ Answer: (a) 0.0228, (b) 0.9772, (c) 0.4987
H	5	A manufacture knows from his expenditure that the resistance of resistor he produces is normal with $\mu = 100$ ohms and $SD = \sigma = 2$ ohms. What percentage of resistors will have resistance between 98 ohms and 102 ohms? Answer: 0.6826
T	6	The average seasonal rainfall in a place is 16 inches with an SD of 4 inches. What is the probability that the rainfall in that place will be between 20 and 24 inches in a year? Answer: 0.1359
H	7	Weights of 500 students of college is normally distributed with $\mu = 95$ lbs. & $\sigma = 7.5$ lbs. Find how many students will have the weight between 100 and 110 lbs. $[P(0 < z < 2) = 0.4772, P(0 < z < 0.67) = 0.2486]$ Answer: 114
C	8	For a random variable having the normal distribution with $\mu = 18.2$ and $\sigma = 1.25$, find the probabilities that it will take on a value less than 16.5, between 16.5 and 18.8. $[P(0 < z < 1.36) = 0.4131, P(0 < z < 0.48) = 0.1844]$ Answer: 0.0869, 0.5974

Unit 3 Probability Distribution

C	9	The compressive strength of the sample of cement can be modelled by normal distribution with mean 6000 kg/cm² and standard deviation 100 kg/cm². (a) What is the probability that a sample strength is less than 6250 kg/cm²? (b) What is probability if sample strength is between 5800 and 5900 kg/cm²? (c) What strength is exceeded by 95% of the samples? [$P(0 < z < 2.5) = 0.4938$, $P(0 < z < 1) = 0.3413$, $P(0 < z < 2) = 0.4772$, $P(0 < z < 1.65) = 0.4505$] Answer: (a) 0.9938, (b) 0.136, (c) 6165
C	10	A sample of 100 dry battery cell tested & found that average life is 12 hours & standard deviation 3 hours. Assuming data to be normally distributed what % of battery cells are expected to have life (a) more than 15 hrs.? (b) less than 6 hrs.? (c) between 10 & 14 hrs.? [$P(0 < z < 1) = 0.3413$, $P(0 < z < 2) = 0.4772$, $P(0 < z < 0.67) = 0.2486$] Answer: (a) 15.87%, (b) 2.27%, (c) 49.72%
H	11	Distribution of height of 1000 soldiers is normal with mean 165 cm & standard deviation 15 cm. How many soldiers are of height (a) less than 138 cm? (b) more than 198 cm? (c) between 138 & 198 cm? [$P(0 < z < 1.8) = 0.4641$, $P(0 < z < 2.2) = 0.4861$] Answer: (a) 36, (b) 14, (c) 950

T	12	In a photographic process, the developing time of prints may be looked upon as a random variable having normal distribution with mean of 16.28 seconds and standard deviation of 0.12 seconds. Find the probability that (a) It will take anywhere from 16.00 to 16.50 sec to develop one of the prints (b) At least 16.20 sec to develop one of the prints (c) At most 16.35 sec to develop one of the prints. [$P(0 < z < 1.83) = 0.4664$, $P(0 < z < 2.33) = 0.4901$] [$P(0 < z < 0.67) = 0.2486$, $P(0 < z < 0.58) = 0.2190$] Answer: (a) 0.9565, (b) 0.7486, (c) 0.7190
H	13	The customer accounts of certain department store have an average balance of 120 Rs. and standard deviation of 40 Rs. Assume that account balances are normally distributed. (a) What proportion of the account is over 150 Rs.? (b) What proportion of account is between 100 & 150 Rs.? (c) What proportion of account is between 60 & 90 Rs.? [$P(0 < z < 0.75) = 0.2734$, $P(0 < z < 0.5) = 0.1915$, $P(0 < z < 1.5) = 0.4332$] Answer: (a) 0.2266, (b) 0.4649, (c) 0.1598
T	14	What is the probability that a standard normal variate Z will be (a) greater than 1.09? (b) less than -1.65? (c) lying between -1 and 1.96? (d) lying between 1.25 and 2.75? Answer: (a) 0.1379, (b) 0.0495, (c) 0.8163, (d) 0.1026
C	15	Assuming that the diameters of 1000 brass plugs taken consecutively from a machine form a normal distribution with mean 0.7515 cm and standard deviation 0.002 cm. Find the number of plugs likely to be rejected if the approved diameter is 0.752 ± 0.004 cm. [$P(0 < z < 1.75) = 0.4599$, $P(0 < z < 2.25) = 0.4878$] Answer: 52

Unit 3 Probability Distribution

T	16	In a company, amount of light bills follows normal distribution with $\sigma = 60$. 11.31% of customers pay bill less than 260. Find average amount of light bill. [$P(z < 1.21) = 0.3869$] Answer: 332.60
C	17	In an examination, minimum 40 marks for passing and 75 marks for distinction are required. In this examination 45% students passed and 9% obtained distinction. Find average marks and standard deviation of this distribution of marks. [$P(0 < z < 0.125) = 0.05$, $P(0 < z < 1.34) = 0.4099$] Answer: 36.40, 28.81

***** End of the Unit *****